

DATA 3464: Fundamentals of Data Processing

Categorical Data Basics

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Topic overview

- Exploring categorical data
- Basic categorical data encoding strategies
- Out of scope: Fancier encoding strategies, like feature hashing and target encoding

Resources used:

- [Feature Engineering Chapter 5](#)
- [Scikit-learn User Guide \(7.3\)](#)

What is categorical data?

- Samples can take on one of several discrete values or groups
 - **Nominal**: no particular order to the groups
 - **Ordinal**: groups relate to each other in a specific order
- Categories can be represented as strings *or* numeric types
 - Domain knowledge is necessary!

Let's take a few minutes to brainstorm some examples

Exploring categorical data

We've already done some of this, but some ideas to consider:

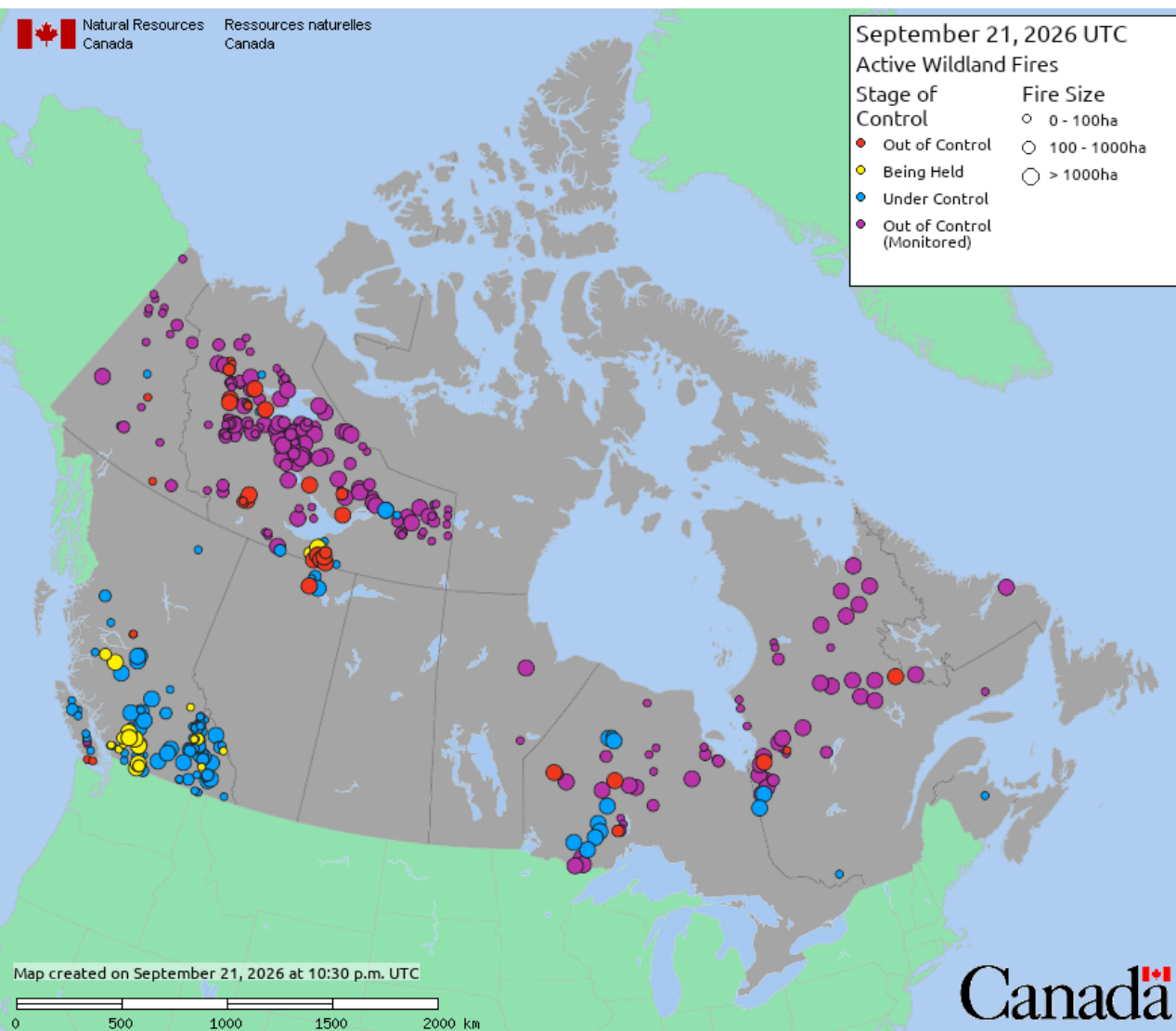
- `pandas.DataFrame.value_counts()` - how many of each category?
- Use category to group, then compute summary stats
- Plot color per category
- Scatter plot with jitter

Representing categorical data

- To do math on categorical data, we need to map to numbers
- Consider:
 - Ordinal or nominal?
 - How many possible categories (cardinality)?
 - Any chance new ones might show up?

How could we encode the examples?

What are the benefits and drawbacks of each method?



Ordinal encoding

Category	Feature
Under Control	0
Being Held	1
*Out of Control (Monitored)	2
Out of Control	3

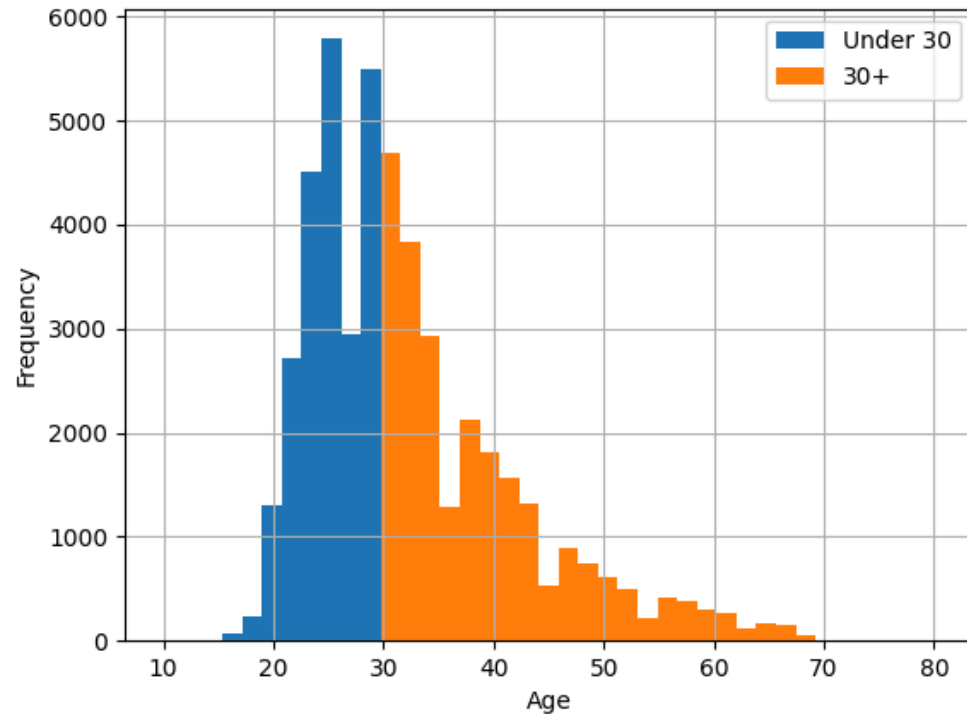
* I am not a wildfire expert, this is my best guess at ordering

Nominal categories: one-hot encoding

- Categories have no natural relationship
- Create k new features from k categories, very sparse matrix

Animal		cat	dog	rabbit
cat	→	1	0	0
dog	→	0	1	0
rabbit	→	0	0	1

From numbers to categories: discretization



- Sometimes numeric data is better represented as categorical
- Still needs encoding strategy
- Can introduce nonlinear relationships
- Like everything, contextual

Advantages/disadvantages of various methods

Ordinal

- + Compact (1 column/feature)
- + Only useful for naturally ordered data
- Imposes (linear) relationship
- Difficult to deal with novel categories

One-hot

- + No assumptions made about relationships
- + Novel features implied by 0
- Memory intensive
- Not good for high cardinality
- Potential collinearity issues

Coming up next

- Lecture: Basic numeric transformations
- Lab: Modelling with categorical data
- Assignment 1 due soon!

Feature Engineering Chapter 5